# Mathematical and Computational Methods

Unit 2: Complex numbers I: algebra and geometry

Unit 1's chain of number systems cannot stop at the reals: $x^2 + 1 = 0$ has no real solution. This unit takes the final step, $\mathbb{R} \subset \mathbb{C}$. We do arithmetic with complex numbers, meet the conjugate and the modulus, picture everything on the *Argand diagram*, and finish with the *polar form*, in which multiplication becomes rotation and scaling.

**Where this fits.** The polar form leans on Unit 1's trigonometry: radians, the values of sin and cos, and the addition formulas. The geometric view of multiplication built here is the springboard for Unit 3: the exponential (Euler) form, De Moivre's theorem and complex roots.

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## Learning objectives

By the end of this unit you will be able to:

- Do complex arithmetic in Cartesian form using the conjugate and modulus.
- Solve real quadratics with complex (conjugate-pair) roots.
- Represent complex numbers on the Argand diagram and add them as vectors.
- Convert between Cartesian and polar form; multiply and divide in polar form as rotation and scaling.

# 1 The algebra of complex numbers

## 1.1 The imaginary unit

To solve $x^2 + 1 = 0$ we need a number whose square is $-1$. No real number qualifies, so we introduce a new one.

**Definition 1.1 (Imaginary unit and complex numbers).** The *imaginary unit* $\mathrm{i}$ is a symbol obeying the single rule

$$\mathrm{i}^2 = -1.$$

A *complex number* is an expression $z = a + b\,\mathrm{i}$ with $a, b \in \mathbb{R}$. The real number $a = \mathrm{Re}(z)$ is the *real part* and $b = \mathrm{Im}(z)$ is the *imaginary part* (itself a real number). The set of all complex numbers is denoted $\mathbb{C}$. Writing $z$ as $a + b\,\mathrm{i}$ is called its *Cartesian form*.

A real number is the special case $b = 0$, so $\mathbb{R} \subset \mathbb{C}$; a number with $a = 0$ and $b \neq 0$, such as $3\mathrm{i}$, is called *purely imaginary*. Two complex numbers are equal exactly when they match in both parts.

**Definition 1.2 (Equality).** $a + b\,\mathrm{i} = c + d\,\mathrm{i}$ if and only if $a = c$ *and* $b = d$. A single complex equation thus encodes *two* real equations.

*Note* (Why $\mathrm{i}^2 = -1$ and not $\mathrm{i} = \sqrt{-1}$). It is tempting to write $\mathrm{i} = \sqrt{-1}$, but this invites contradiction: blindly applying $\sqrt{x}\sqrt{y} = \sqrt{xy}$ would give $1 = \sqrt{1} = \sqrt{(-1)(-1)} = \sqrt{-1}\,\sqrt{-1} = \mathrm{i} \cdot \mathrm{i} = -1$. The rule $\sqrt{x}\sqrt{y} = \sqrt{xy}$ holds only for non-negative reals. We therefore *define* $\mathrm{i}$ by the property $\mathrm{i}^2 = -1$ and never manipulate $\sqrt{-1}$ directly.

Because $\mathrm{i}^2 = -1$, the higher powers of $\mathrm{i}$ cycle with period four. From $\mathrm{i}^2 = -1$ we get $\mathrm{i}^3 = \mathrm{i}^2 \cdot \mathrm{i} = -\mathrm{i}$ and $\mathrm{i}^4 = (\mathrm{i}^2)^2 = 1$, after which the pattern repeats:

$$\mathrm{i}^0 = 1, \quad \mathrm{i}^1 = \mathrm{i}, \quad \mathrm{i}^2 = -1, \quad \mathrm{i}^3 = -\mathrm{i}, \quad \mathrm{i}^4 = 1, \quad \mathrm{i}^5 = \mathrm{i}, \ \ldots$$

**Example 1.1** ***(Powers of the imaginary unit).*** To evaluate a power of $\mathrm{i}$, divide the exponent by 4 and keep only the remainder (this is called *reducing it modulo 4*). Since $2025 = 4 \cdot 506 + 1$,

$$\mathrm{i}^{2025} = \mathrm{i}^{1} = \mathrm{i}.$$

Negative powers follow from $\mathrm{i}^{-1} = \dfrac{1}{\mathrm{i}} = \dfrac{\mathrm{i}}{\mathrm{i}^2} = -\mathrm{i}$.

## 1.2 Addition, subtraction and multiplication

We extend the arithmetic of $\mathbb{R}$ to $\mathbb{C}$ by requiring the familiar laws—commutativity, associativity and distributivity—to keep holding, together with the single new rule $\mathrm{i}^2 = -1$. These requirements leave no freedom: they *force* the formulas below, which we therefore take as the definition of the three operations.

**Definition 1.3 (The three operations).** For $z = a + b\,\mathrm{i}$ and $w = c + d\,\mathrm{i}$,

$$\begin{aligned} z + w &= (a + c) + (b + d)\,\mathrm{i}, \\ z - w &= (a - c) + (b - d)\,\mathrm{i}, \\ z\,w &= (ac - bd) + (ad + bc)\,\mathrm{i}. \end{aligned}$$

Addition and subtraction are part by part. The product is just the bracket expansion, with $\mathrm{i}^2$ replaced by $-1$:

$$(a + b\mathrm{i})(c + d\mathrm{i}) = ac + ad\,\mathrm{i} + bc\,\mathrm{i} + bd\,\mathrm{i}^2 = (ac - bd) + (ad + bc)\,\mathrm{i}.$$

**Example 1.2 *(Cartesian arithmetic).*** Let $z = 2 + 3\mathrm{i}$ and $w = 1 - 4\mathrm{i}$. Then

$$z + w = 3 - \mathrm{i}, \qquad z - w = 1 + 7\mathrm{i},$$

and, using the product rule with $a = 2,\ b = 3,\ c = 1,\ d = -4$,

$$z\,w = (2 \cdot 1 - 3 \cdot (-4)) + (2 \cdot (-4) + 3 \cdot 1)\,\mathrm{i} = (2 + 12) + (-8 + 3)\,\mathrm{i} = 14 - 5\mathrm{i}.$$

Equality of complex numbers turns one complex equation into two real ones. To find the real numbers $x, y$ with $(x + y\,\mathrm{i})(1 + \mathrm{i}) = 3 + \mathrm{i}$, expand the left side:

$$(x - y) + (x + y)\,\mathrm{i} = 3 + \mathrm{i}.$$

Matching real and imaginary parts gives $x - y = 3$ and $x + y = 1$, so $x = 2$, $y = -1$. Check: $(2 - \mathrm{i})(1 + \mathrm{i}) = 2 + 2\mathrm{i} - \mathrm{i} - \mathrm{i}^2 = 3 + \mathrm{i}$.

*Remark.* The square of a complex number need not be a non-negative real number—indeed it need not be real at all. For example $(2 + 3\mathrm{i})^2 = 4 + 12\mathrm{i} + 9\mathrm{i}^2 = -5 + 12\mathrm{i}$. This is why the real-number test "a square is $\geq 0$" has no analogue in $\mathbb{C}$.

## 1.3 The complex conjugate

**Definition 1.4 (Conjugate).** The *complex conjugate* of $z = a + b\,\mathrm{i}$ is $\overline{z} = a - b\,\mathrm{i}$: the sign of the imaginary part is reversed.

Conjugation interacts cleanly with every operation. The following are checked by a one-line calculation each.

**Proposition 1.1 (Properties of the conjugate).** For all $z, w \in \mathbb{C}$, writing $z = a + b\mathrm{i}$,

$$\overline{\overline{z}} = z, \qquad \overline{z + w} = \overline{z} + \overline{w}, \qquad \overline{z\,w} = \overline{z}\,\overline{w},$$

$$z + \overline{z} = 2\,\mathrm{Re}(z), \qquad z - \overline{z} = 2\,\mathrm{Im}(z)\,\mathrm{i}, \qquad z\,\overline{z} = a^2 + b^2.$$

Moreover $z$ is real $\iff \overline{z} = z$, and $\overline{z} = -z$ exactly when $\mathrm{Re}(z) = 0$ (that is, $z$ purely imaginary or zero).

The last identity is the most useful: the product of a number with its conjugate is always a *non-negative real number*,

$$z\,\overline{z} = (a + b\mathrm{i})(a - b\mathrm{i}) = a^2 - (b\mathrm{i})^2 = a^2 + b^2.$$

This single fact powers both the modulus and division below.

## 1.4 The modulus

**Definition 1.5 (Modulus).** The *modulus* (or absolute value) of $z = a + b\,\mathrm{i}$ is the non-negative real number

$$|z| = \sqrt{a^2 + b^2}, \qquad \text{so that} \qquad |z|^2 = z\,\overline{z}.$$

**Proposition 1.2 (Properties of the modulus).** For all $z, w \in \mathbb{C}$,

$$|z| \geq 0 \text{ with } |z| = 0 \iff z = 0, \quad |\overline{z}| = |z|\,, \quad |z\,w| = |z|\;|w|\,,$$

and the *triangle inequality* $|z + w| \leq |z| + |w|$.

**Example 1.3** ***(Computing a modulus).*** For $z = 12 - 5\mathrm{i}$,

$$|z| = \sqrt{12^2 + (-5)^2} = \sqrt{144 + 25} = \sqrt{169} = 13.$$

## 1.5 Why the modulus behaves well

Two of the properties just listed—the multiplicativity $|zw| = |z|\,|w|$ and the triangle inequality—are worth proving, not merely asserting. Both run on the single identity $z\,\overline{z} = |z|^2$ flagged above as the most useful one.

**Proposition 1.3 (Multiplicativity of the modulus).** For all $z, w \in \mathbb{C}$, $|z\,w| = |z|\;|w|$.

*Proof.* Square the left-hand side and use $|u|^2 = u\,\overline{u}$ together with $\overline{z\,w} = \overline{z}\,\overline{w}$ (from the conjugate properties):

$$|z\,w|^2 = (z\,w)\,\overline{z\,w} = (z\,w)(\overline{z}\,\overline{w}) = (z\,\overline{z})(w\,\overline{w}) = |z|^2\;|w|^2\,.$$

Both $|z\,w|$ and $|z|\;|w|$ are non-negative, so taking square roots gives $|z\,w| = |z|\;|w|$. ∎

For the triangle inequality we need one small fact: for any $\zeta = p + q\,\mathrm{i}$,

$$\operatorname{Re}\zeta = p \leq \sqrt{p^2 + q^2} = |\zeta|\,.$$

**Proposition 1.4 (Triangle inequality).** For all $z, w \in \mathbb{C}$, $|z + w| \leq |z| + |w|$, with equality exactly when one of $z$, $w$ is a non-negative real multiple of the other.

*Proof.* Expand $|z + w|^2$ with $|u|^2 = u\,\overline{u}$ and $\overline{z + w} = \overline{z} + \overline{w}$:

$$|z + w|^2 = (z + w)(\overline{z} + \overline{w}) = z\overline{z} + w\overline{w} + (z\overline{w} + \overline{z}\,w) = |z|^2 + |w|^2 + 2\operatorname{Re}(z\overline{w}),$$

since $\overline{z}\,w = \overline{z\overline{w}}$ and $\zeta + \overline{\zeta} = 2\operatorname{Re}\zeta$. Bounding the middle term by the fact above and then by multiplicativity ($|z\overline{w}| = |z|\,|\overline{w}| = |z|\,|w|$),

$$|z + w|^2 \leq |z|^2 + 2\,|z|\,|w| + |w|^2 = \big(|z| + |w|\big)^2.$$

Taking non-negative square roots gives $|z + w| \leq |z| + |w|$. Equality holds exactly when $\operatorname{Re}(z\overline{w}) = |z\overline{w}|$, that is, when $z\overline{w} = t$ is a real number $t \geq 0$. For $w \neq 0$, multiplying by $w$ turns this into $z = \frac{t}{|w|^2}\,w$, a non-negative real multiple of $w$. (On the Argand diagram of Section 2 this says that the two arrows point the same way.) ∎

*Remark.* The triangle inequality is the statement that "the straight path is shortest": the side $|z+w|$ of a triangle cannot exceed the sum of the other two sides $|z|$ and $|w|$. The same inequality, proved by the same expand-and-bound idea, holds for the length (norm) of vectors in $\mathbb{R}^n$, which Unit 4 introduces.

## 1.6 Division

Division is made possible by the fact that $w\overline{w} = |w|^2$ is real: to divide by $w$, multiply top and bottom by $\overline{w}$, which clears the imaginary part from the denominator.

**Theorem 1.1 (Division).** For $z = a + b\,\mathrm{i}$ and $w = c + d\,\mathrm{i} \neq 0$,

$$\frac{z}{w} = \frac{z\,\overline{w}}{w\,\overline{w}} = \frac{z\,\overline{w}}{|w|^2} = \frac{(ac+bd) + (bc-ad)\,\mathrm{i}}{c^2+d^2}.$$

In particular the reciprocal is $\dfrac{1}{z} = \dfrac{\overline{z}}{|z|^2}$.

With division in hand, the modulus of a quotient follows from multiplicativity.

**Corollary 1.1 (Modulus of a quotient).** If $w \neq 0$ then $\left|\frac{z}{w}\right| = \dfrac{|z|}{|w|}$.

*Proof.* Apply multiplicativity of the modulus to the identity $\frac{z}{w} \cdot w = z$: it gives $\left|\frac{z}{w}\right|\,|w| = |z|$, and dividing by $|w| > 0$ finishes. ∎

**Example 1.4** ***(Dividing in Cartesian form).*** With $z = 2+3\mathrm{i}$ and $w = 1-4\mathrm{i}$ as before, multiply top and bottom by the conjugate $\overline{w} = 1+4\mathrm{i}$ of the denominator:

$$\frac{2+3\mathrm{i}}{1-4\mathrm{i}} = \frac{(2+3\mathrm{i})(1+4\mathrm{i})}{(1-4\mathrm{i})(1+4\mathrm{i})} = \frac{2+8\mathrm{i}+3\mathrm{i}+12\,\mathrm{i}^2}{1^2+4^2} = \frac{-10+11\mathrm{i}}{17} = -\tfrac{10}{17} + \tfrac{11}{17}\,\mathrm{i}.$$

A quotient can also come out real. For $\dfrac{-3+4\mathrm{i}}{6-8\mathrm{i}}$, multiply by $6+8\mathrm{i}$:

$$\frac{-3+4\mathrm{i}}{6-8\mathrm{i}} = \frac{(-3+4\mathrm{i})(6+8\mathrm{i})}{(6-8\mathrm{i})(6+8\mathrm{i})} = \frac{-18-24\mathrm{i}+24\mathrm{i}+32\,\mathrm{i}^2}{6^2+8^2} = \frac{-50}{100} = -\tfrac{1}{2},$$

because $-3+4\mathrm{i} = -\frac{1}{2}(6-8\mathrm{i})$: the numerator is a real multiple of the denominator.

*Note* (A common error). Division is *not* done part by part: $\dfrac{a+b\mathrm{i}}{c+d\mathrm{i}} \neq \dfrac{a}{c} + \dfrac{b}{d}\,\mathrm{i}$. Always multiply by the conjugate of the denominator.

## 1.7 Quadratic equations with complex roots

We can now keep the promise made at the start of the unit.

**Example 1.5** ***(The equation that motivated $\mathbb{C}$).*** $x^2 + 1 = 0$ means $x^2 = -1$. Since $\mathrm{i}^2 = -1$ and $(-\mathrm{i})^2 = -1$, the two solutions are $x = \pm\mathrm{i}$. The equation that had no real root has exactly two complex ones.

More generally, a real quadratic $ax^2 + bx + c = 0$ whose discriminant $b^2 - 4ac$ is negative has no real roots, but completing the square produces a pair of complex ones. We can do this once and for all.

**Theorem 1.2 (The quadratic formula, negative-discriminant case).** Let $a, b, c \in \mathbb{R}$ with $a \neq 0$, and suppose the discriminant $D = b^2 - 4ac$ is *negative*. Then $ax^2 + bx + c = 0$ has exactly the two complex roots

$$x = \frac{-b \pm \mathrm{i}\sqrt{4ac - b^2}}{2a},$$

which form a conjugate pair with real part $-\dfrac{b}{2a}$ and imaginary parts $\pm\dfrac{\sqrt{4ac - b^2}}{2a}$.

*Proof.* Complete the square:

$$ax^2 + bx + c = a\left(x + \tfrac{b}{2a}\right)^2 + c - \tfrac{b^2}{4a} = a\left(x + \tfrac{b}{2a}\right)^2 - \tfrac{D}{4a},$$

so the equation becomes $\left(x + \frac{b}{2a}\right)^2 = \dfrac{D}{4a^2}$. When $D < 0$ the right-hand side is negative, and because $\mathrm{i}^2 = -1$ it is a square after all:

$$\frac{D}{4a^2} = -\frac{4ac - b^2}{4a^2} = \left(\frac{\mathrm{i}\sqrt{4ac - b^2}}{2a}\right)^2,$$

where $4ac - b^2 > 0$, so only an ordinary real square root appears. Hence $x + \frac{b}{2a} = \pm\dfrac{\mathrm{i}\sqrt{4ac - b^2}}{2a}$, and rearranging yields the stated formula. ∎

*Note* (The same formula you already know). This is not a new formula but the familiar $x = \left(-b \pm \sqrt{D}\right)/2a$ written so as to keep the imaginary unit outside the square root. When $D \geq 0$ the roots are real; when $D < 0$ the factor i surfaces and the roots are a conjugate pair. The case $D = 0$ gives the single repeated real root $-b/2a$.

**Example 1.6** ***(A quadratic with complex roots).*** Solve $x^2 + 2x + 10 = 0$. Completing the square,

$$(x + 1)^2 + 9 = 0 \iff (x + 1)^2 = -9 \iff x + 1 = \pm 3\mathrm{i},$$

using $(3\mathrm{i})^2 = 9\,\mathrm{i}^2 = -9$. Hence $x = -1 \pm 3\mathrm{i}$.

**Example 1.7** ***(Applying the formula).*** Solve $2x^2 + 2x + 5 = 0$. Here $a = 2$, $b = 2$, $c = 5$, so $D = b^2 - 4ac = 4 - 40 = -36 < 0$ and $4ac - b^2 = 36$. The formula gives

$$x = \frac{-2 \pm \mathrm{i}\sqrt{36}}{2 \cdot 2} = \frac{-2 \pm 6\,\mathrm{i}}{4} = -\tfrac{1}{2} \pm \tfrac{3}{2}\mathrm{i}.$$

As a check, the conjugate pair multiplies back to a real factor: $\left(x+\frac{1}{2}\right)^2+\frac{9}{4}=x^2+x+\frac{5}{2}$, which is $\frac{1}{2}(2x^2+2x+5)$.

**Proposition 1.5 (Conjugate-pair theorem).** Let $p(x)=c_nx^n+\cdots+c_1x+c_0$ be a polynomial with *real* coefficients $c_0,\dots,c_n\in\mathbb{R}$. If $\alpha\in\mathbb{C}$ satisfies $p(\alpha)=0$, then $p(\overline{\alpha})=0$ as well: the non-real roots of a real polynomial occur in conjugate pairs.

*Proof.* Conjugation respects sums and products and fixes real numbers, so $\overline{c_k}=c_k$ and $\overline{\alpha^k}=(\overline{\alpha})^k$. Hence

$$\overline{p(\alpha)}=\overline{\sum_{k=0}^{n}c_k\alpha^k}=\sum_{k=0}^{n}\overline{c_k}\,\overline{\alpha}^{\,k}=\sum_{k=0}^{n}c_k(\overline{\alpha})^k=p(\overline{\alpha}).$$

Since $p(\alpha)=0$, this gives $p(\overline{\alpha})=\overline{0}=0$. ∎

Pairing a non-real root $\alpha$ with $\overline{\alpha}$ explains why such roots leave a *real* quadratic factor behind:

$$(x-\alpha)(x-\overline{\alpha})=x^2-(\alpha+\overline{\alpha})x+\alpha\overline{\alpha}=x^2-2\,\mathrm{Re}(\alpha)\,x+|\alpha|^2\,,$$

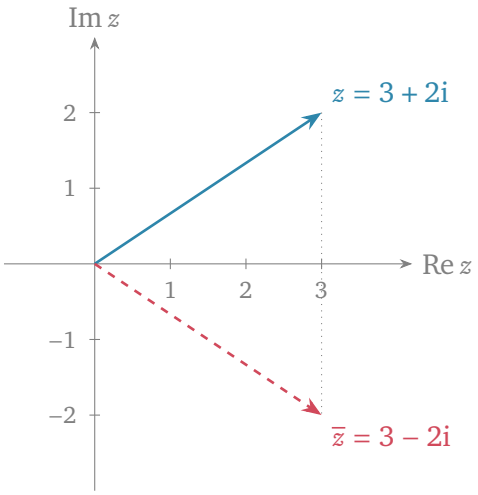
whose coefficients are real. For $\alpha=-1+3\mathrm{i}$ this reads $x^2+2x+10$, recovering the quadratic above.

*Remark* (The fundamental theorem of algebra). Every degree-$n$ polynomial—real or complex coefficients—has exactly $n$ roots in $\mathbb{C}$ when these are *counted with multiplicity* (so a repeated root such as the double root of $x^2-2x+1=(x-1)^2$ counts twice). This is the fundamental theorem of algebra, which we return to in Unit 3.

## 2 The Argand diagram

### 2.1 Complex numbers as points and vectors

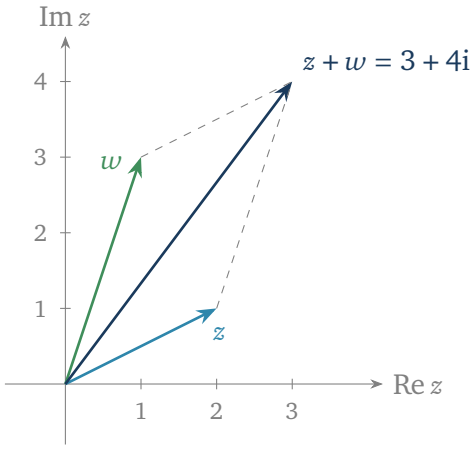
Because a complex number $z=a+b\,\mathrm{i}$ is determined by the real pair $(a,b)$, it can be drawn as a point—or as the arrow from the origin to that point—in the plane. The horizontal axis carries the real part and the vertical axis the imaginary part; the plane used this way is the *Argand diagram* (or complex plane).



The picture immediately makes the modulus and the conjugate geometric: $|z|$ is the *length* of the arrow (by Pythagoras), and $\overline{z}$ is the *reflection of $z$ in the real axis*. Likewise $-z$ is the reflection of $z$ through the origin.

## 2.2 Addition and subtraction as vectors

Since complex addition adds real and imaginary parts separately, it behaves exactly like vector addition in the plane $\mathbb{R}^2$ (pairs of real numbers, the subject of Unit 4): $z + w$ is found by the tip-to-tail (parallelogram) rule.




Subtraction has an equally useful reading: $z - w$ is the vector *from $w$ to $z$*, so $|z - w|$ is the *distance between the two points* $z$ and $w$ in the plane. This turns many geometric statements into short algebraic ones.

**Example 2.1** ***(Sum and distance).*** For $z = 2 + \mathrm{i}$ and $w = 1 + 3\mathrm{i}$ we have $z + w = 3 + 4\mathrm{i}$, drawn above. The distance between the points is

$$|z - w| = |(2 - 1) + (1 - 3)\mathrm{i}| = |1 - 2\mathrm{i}| = \sqrt{1^2 + (-2)^2} = \sqrt{5}.$$

**Example 2.2** ***(A circle, a bisector and a disc).*** Reading $|z - c|$ as the distance from $z$ to the point $c$ turns a condition on $z$ into a set of points. The equation $|z - 3 - \mathrm{i}| = 2$ reads $|z - (3 + \mathrm{i})| = 2$: the points at distance 2 from $3 + \mathrm{i}$, i.e. the *circle* with centre $3 + \mathrm{i}$ and radius 2. In general $|z - c| = r$ is the circle with centre $c$ and radius $r$.

The equation $|z - 2| = |z - 2\mathrm{i}|$ collects the points as far from 2 as from $2\mathrm{i}$: the *perpendicular bisector* of the segment joining them. To confirm it, put $z = x + y\,\mathrm{i}$ and square both sides, which are non-negative:

$$(x - 2)^2 + y^2 = x^2 + (y - 2)^2$$

$$-4x + 4 = -4y + 4,$$

so $y = x$: the line through the midpoint $1 + \mathrm{i}$, at right angles to the segment. In general $|z - a| = |z - b|$ is the perpendicular bisector of the segment from $a$ to $b$.

The inequality $|z + \mathrm{i}| < 2$ reads $|z - (-\mathrm{i})| < 2$: the points at distance *less than* 2 from $-\mathrm{i}$, the *open disc* with centre $-\mathrm{i}$ and radius 2. The origin lies inside, as $|0 + \mathrm{i}| = 1 < 2$, but the

boundary circle $|z + \mathrm{i}| = 2$ is excluded (for instance $z = \mathrm{i}$, with $|2\mathrm{i}| = 2$), so a sketch draws it dashed.

*Remark* (A data-science reading). Identifying $z = a + b\,\mathrm{i}$ with the point $(a, b)$, the modulus $|z - w|$ is exactly the *Euclidean distance* between two data points in the plane—the same distance that measures how alike two records are in machine learning. The plane-geometry of complex numbers is therefore the two-dimensional case of the $\mathbb{R}^n$ distance developed in Unit 4; multiplication, which we turn to next, adds a rotation that has no real-number counterpart and reappears as the phase of a signal in Unit 3.

### 2.3 Conjugation, modulus and the absence of order

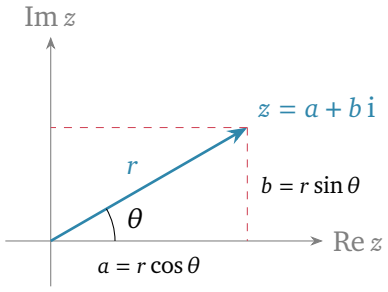
The Argand diagram gives $|z|$, $\overline{z}$ and addition a geometric meaning. One familiar feature of the real numbers, however, is lost.

*Remark* (The complex numbers cannot be ordered). On the real line any two numbers satisfy exactly one of $<, =, >$, and the order obeys two rules that make it useful: adding the same number to both sides keeps an inequality, and the product of two positive numbers is positive. No ordering of $\mathbb{C}$ can obey both. If one did, then any $x \neq 0$ would have $x > 0$ or $-x > 0$, and either way the second rule gives $x^2 = (-x)^2 > 0$; applying this to $x = \mathrm{i}$ gives $\mathrm{i}^2 = -1 > 0$, while applying it to $x = 1$ gives $1 = 1^2 > 0$, i.e. $-1 < 0$. These contradict one another. So expressions like "$z < w$" are meaningless for non-real complex numbers—only the real quantities $|z|$, $\mathrm{Re}(z)$, $\mathrm{Im}(z)$ can be compared.

## 3 The polar form

### 3.1 Modulus and argument

A point in the plane can be located by Cartesian coordinates $(a, b)$ or, just as well, by how *far* it is from the origin and in which *direction*. For a nonzero $z$, write $r = |z|$ for the distance and let $\theta$ be the angle the arrow makes with the positive real axis, measured anticlockwise in radians (the *argument* of $z$). Reading off the right-angled triangle—the radians and standard values of Unit 1, now read round a circle—gives




**Definition 3.1 (Polar form).** Every nonzero $z \in \mathbb{C}$ can be written in *polar form*

$$z = r(\cos\theta + \mathrm{i}\sin\theta), \qquad r = |z| = \sqrt{a^2 + b^2} > 0, \qquad \theta = \arg(z),$$

where $a = r\cos\theta$ and $b = r\sin\theta$. The angle $\theta$ is only determined up to whole turns: if $\theta$ is

one argument of $z$, then so is $\theta + 2\pi k$ for every $k \in \mathbb{Z}$, and all describe the same number; $\arg(z)$ denotes any one of them. The unique value in $(-\pi, \pi]$ is the *principal argument* $\operatorname{Arg}(z)$. (For $z = 0$ the modulus is 0 and the argument is undefined.)

The radius is found from $r = \sqrt{a^2 + b^2}$, and the angle from $\tan\theta = b/a$—but the tangent alone does not fix $\theta$, since tan has period $\pi$ (Unit 1). The Argand diagram decides which quadrant $\theta$ lies in. (When $a = 0$ the tangent is undefined: the point is on the imaginary axis, and $\theta = \frac{\pi}{2}$ if $b > 0$, $\theta = -\frac{\pi}{2}$ if $b < 0$.) Since $\overline{z}$ is the reflection of $z$ in the real axis, it has the same modulus and the opposite angle: $\arg\overline{z} = -\arg z$.

*Note* (Mind the quadrant). Let $\alpha = \arctan|b/a|$ be the *reference angle*, the acute angle between the arrow and the real axis. Sketch the point, then read off the principal argument:

| quadrant | first | second | third | fourth |
|---|---|---|---|---|
| signs of $(a, b)$ | $(+,+)$ | $(-,+)$ | $(-,-)$ | $(+,-)$ |
| $\operatorname{Arg}(z)$ | $\alpha$ | $\pi - \alpha$ | $-(\pi - \alpha)$ | $-\alpha$ |

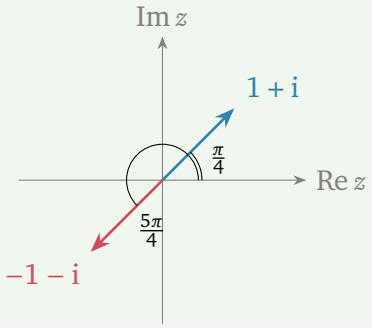
A calculator's $\arctan(b/a)$ agrees with this only in the first and fourth quadrants.

## 3.2 Converting between Cartesian and polar form

**Example 3.1** ***(Cartesian → polar, with quadrant care).*** Convert $z = 1 + \mathrm{i}$ and $w = -1 - \mathrm{i}$. Both have $r = \sqrt{1^2 + 1^2} = \sqrt{2}$ and $\tan\theta = 1$, but the diagram below separates them: $z$ sits in the first quadrant at $\theta = \frac{\pi}{4}$, while $w$ sits in the third at $\theta = \frac{5\pi}{4}$. Hence

$$z = \sqrt{2}\left(\cos\tfrac{\pi}{4} + \mathrm{i}\sin\tfrac{\pi}{4}\right), \qquad w = \sqrt{2}\left(\cos\tfrac{5\pi}{4} + \mathrm{i}\sin\tfrac{5\pi}{4}\right).$$

Here $\frac{5\pi}{4}$ is *an* argument of $w$; its principal value is $\operatorname{Arg}(w) = -\frac{3\pi}{4}$.




The two numbers above both make a 45° angle with the real axis; a less symmetric number exercises the full recipe.

**Example 3.2** ***(Cartesian → polar in the second quadrant).*** Convert $z = -\sqrt{3}+\mathrm{i}$. The modulus is $r = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{3+1} = 2$. The reference angle comes from the side lengths, $\arctan\dfrac{1}{\sqrt{3}} = \dfrac{\pi}{6}$. The point lies in the *second* quadrant (real part negative, imaginary part positive), and the reference angle is the one made with the negative real axis, so the argument—always

measured from the *positive* real axis—is $\pi$ minus it:

$$\operatorname{Arg}(z) = \pi - \tfrac{\pi}{6} = \tfrac{5\pi}{6}, \qquad z = 2\Big(\cos\tfrac{5\pi}{6} + \mathrm{i}\sin\tfrac{5\pi}{6}\Big).$$

Feeding the raw ratio $b/a = 1/(-\sqrt{3})$ to a calculator returns $\arctan(-1/\sqrt{3}) = -\tfrac{\pi}{6}$, a *fourth*-quadrant angle—the wrong place. The sketch, not the tangent, fixes the quadrant.

Conversion the other way is just evaluation.

**Example 3.3** ***(Polar → Cartesian).*** For $z = 4\big(\cos\tfrac{\pi}{3} + \mathrm{i}\sin\tfrac{\pi}{3}\big)$, read off the trigonometric values from Unit 1:

$$z = 4\Big(\tfrac{1}{2} + \mathrm{i}\,\tfrac{\sqrt{3}}{2}\Big) = 2 + 2\sqrt{3}\,\mathrm{i}.$$

## 3.3 Multiplication and division in polar form

The polar form turns multiplication—awkward in Cartesian coordinates—into a simple rule, and reveals its geometric meaning.

**Theorem 3.1 (Products and quotients in polar form).** If $z = r(\cos\theta + \mathrm{i}\sin\theta)$ and $w = s(\cos\varphi + \mathrm{i}\sin\varphi)$ with $s \neq 0$, then

$$z\,w = rs\big(\cos(\theta+\varphi) + \mathrm{i}\sin(\theta+\varphi)\big), \qquad \frac{z}{w} = \frac{r}{s}\big(\cos(\theta-\varphi) + \mathrm{i}\sin(\theta-\varphi)\big).$$

*Proof.* Multiply out and group, then use the addition formulas for cos and sin from Unit 1:

$$\begin{aligned} z\,w &= rs\big[(\cos\theta\cos\varphi - \sin\theta\sin\varphi) + \mathrm{i}(\sin\theta\cos\varphi + \cos\theta\sin\varphi)\big] \\ &= rs\big[\cos(\theta+\varphi) + \mathrm{i}\sin(\theta+\varphi)\big]. \end{aligned}$$

For the quotient, first note that $\frac{1}{w} = \frac{1}{s}\big(\cos(-\varphi) + \mathrm{i}\sin(-\varphi)\big)$, because $s(\cos\varphi + \mathrm{i}\sin\varphi)\cdot\frac{1}{s}(\cos\varphi - \mathrm{i}\sin\varphi) = \cos^2\varphi + \sin^2\varphi = 1$; then multiply $z$ by $\frac{1}{w}$ using the product rule just proved. ∎

*Note* (Multiplication = scaling and rotation). To multiply two complex numbers, *multiply the moduli and add the arguments*; to divide, *divide the moduli and subtract the arguments*. Geometrically, multiplying by $w$ scales every arrow by $|w|$ and rotates it by $\arg(w)$.

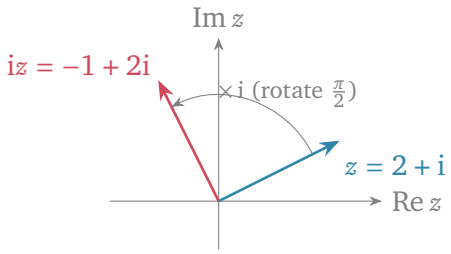
**Example 3.4** ***(A product in polar form).*** Let $z = 2\big(\cos\tfrac{\pi}{6} + \mathrm{i}\sin\tfrac{\pi}{6}\big)$ and $w = 3\big(\cos\tfrac{\pi}{3} + \mathrm{i}\sin\tfrac{\pi}{3}\big)$. Then

$$z\,w = 2\cdot 3\Big(\cos\big(\tfrac{\pi}{6}+\tfrac{\pi}{3}\big) + \mathrm{i}\sin\big(\tfrac{\pi}{6}+\tfrac{\pi}{3}\big)\Big) = 6\Big(\cos\tfrac{\pi}{2} + \mathrm{i}\sin\tfrac{\pi}{2}\Big) = 6\mathrm{i}.$$

**Example 3.5** ***(A quotient in polar form).*** With the same $z$ and $w$, divide the moduli and subtract the arguments:

$$\frac{z}{w} = \frac{2}{3}\Big(\cos\big(\tfrac{\pi}{6}-\tfrac{\pi}{3}\big) + \mathrm{i}\sin\big(\tfrac{\pi}{6}-\tfrac{\pi}{3}\big)\Big) = \frac{2}{3}\Big(\cos\big(-\tfrac{\pi}{6}\big) + \mathrm{i}\sin\big(-\tfrac{\pi}{6}\big)\Big) = \frac{2}{3}\Big(\tfrac{\sqrt{3}}{2} - \tfrac{1}{2}\mathrm{i}\Big) = \frac{\sqrt{3}-\mathrm{i}}{3}.$$

The cleanest special case is multiplication by i itself. Since $\mathrm{i} = \cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}$ has modulus 1 and argument $\frac{\pi}{2}$, multiplying any $z$ by i leaves its length unchanged and turns it through a quarter turn anticlockwise.




**Example 3.6** ***(Multiplication by*** i ***rotates by a quarter turn).*** With $z = 2 + \mathrm{i}$, $\mathrm{i}z = \mathrm{i}(2 + \mathrm{i}) = 2\mathrm{i} + \mathrm{i}^2 = -1 + 2\mathrm{i}$. The modulus is unchanged, $|z| = |\mathrm{i}z| = \sqrt{5}$, and the argument has increased by $\frac{\pi}{2}$, as the figure shows.

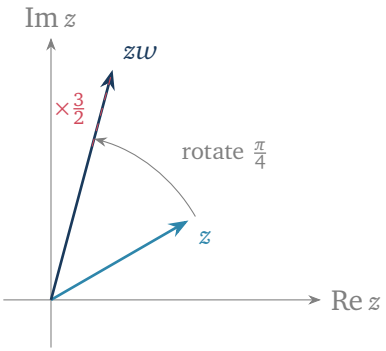
## 3.4 A general product as rotation and scaling

Multiplying by i is the special case $|w| = 1$, $\arg w = \frac{\pi}{2}$: a pure quarter-turn. The next figure shows a $w$ that is neither a pure rotation nor a pure scaling.

**Example 3.7** ***(A general product, geometrically).*** Take $z = 2\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)$ and multiply by $w$ with $|w| = \frac{3}{2}$ and $\arg w = \frac{\pi}{4}$. By the product rule the moduli multiply and the arguments add:

$$|zw| = 2 \cdot \tfrac{3}{2} = 3, \qquad \arg(zw) = \tfrac{\pi}{6} + \tfrac{\pi}{4} = \tfrac{5\pi}{12}.$$

So the arrow for $z$ is rotated anticlockwise through $\arg w = \frac{\pi}{4}$ and stretched by the factor $|w| = \frac{3}{2}$, landing on $zw$ at modulus 3 and argument $\frac{5\pi}{12}$.




The reading reverses for division: dividing by $w$ rotates by $-\arg w$ and scales by $1/|w|$, undoing both moves. In engineering language, multiplying a signal by $w$ applies a phase shift $\arg w$ and a gain $|w|$ at once; the rotation half is exactly the plane rotation that reappears as a rotation matrix in Unit 5.

## Summary

| Concept | Key idea |
|---|---|
| Imaginary unit | $\mathrm{i}^2 = -1$; complex number $z = a + b\,\mathrm{i}$ with $\mathrm{Re}(z) = a$, $\mathrm{Im}(z) = b$; powers of i cycle with period 4. |
| Arithmetic | Add/subtract part by part; multiply by expanding and using $\mathrm{i}^2 = -1$: $zw = (ac - bd) + (ad + bc)\mathrm{i}$. |
| Conjugate | $\overline{z} = a - b\,\mathrm{i}$; $z\,\overline{z} = a^2 + b^2 = \lvert z\rvert^2$ is real and non-negative. |
| Modulus | $\lvert z\rvert = \sqrt{a^2 + b^2}$; multiplicative, $\lvert zw\rvert = \lvert z\rvert\,\lvert w\rvert$; triangle inequality. |
| Division | Multiply top and bottom by $\overline{w}$: $z/w = z\,\overline{w}/\lvert w\rvert^2$. |
| Quadratics | If $D = b^2 - 4ac < 0$, $x = \left(-b \pm \mathrm{i}\sqrt{4ac - b^2}\right)/2a$, a conjugate pair; non-real roots of a real polynomial come in conjugate pairs. |
| Argand diagram | $z \leftrightarrow$ point/arrow $(a, b)$; $\lvert z\rvert$ is its length, $\overline{z}$ its reflection in the real axis; addition is the parallelogram rule and $\lvert z - w\rvert$ is a distance, so $\lvert z - c\rvert = r$ is a circle and $\lvert z - a\rvert = \lvert z - b\rvert$ a perpendicular bisector. |
| No order | $\mathbb{C}$ admits no ordering compatible with arithmetic. |
| Polar form | $z = r(\cos\theta + \mathrm{i}\sin\theta)$ with $r = \lvert z\rvert$, $\theta = \arg(z)$ (fix the quadrant from the diagram; principal value $\mathrm{Arg}(z) \in (-\pi, \pi]$). |
| Polar arithmetic | For $zw$ multiply the moduli and add the arguments; for $z/w$ divide the moduli and subtract them. Multiplication = scaling + rotation; $\times\mathrm{i}$ is a quarter turn. |

*Next unit: the exponential (Euler) form $z = r\,\mathrm{e}^{\mathrm{i}\theta}$, De Moivre's theorem, and the powers and roots of complex numbers.*